Counterexample to Conjecture 9305: Local Weak Limits Need Not Be Supported on Transitive Graphs

gaochengzhecpu

The claim being disproved. The source asserts that local limits admit a decomposition into extreme points whose supports consist of transitive graphs. Here "transitive" means vertex-transitive: the automorphism group acts transitively on the vertices of the underlying graph. We disprove the support assertion. In fact the counterexample cannot be a mixture of *any* laws concentrated on vertex-transitive graphs, regardless of extremality. The source places no connectedness or infinitude requirement on the approximating finite graphs or their limiting root components.

The rooted graph space. Let $\mathcal{G}_*$ consist of rooted isomorphism classes of countable, connected, locally finite simple graphs. The radius-r ball is the induced graph on vertices joined to the root by a walk of length at most r; isomorphisms of balls must preserve the root. Equip $\mathcal{G}_*$ with its local sigma algebra generated by these ball-isomorphism cylinders. The uniformly rooted law of a finite graph means: choose one of its actual vertices uniformly, take its actual connected component, and retain the root. Thus disconnected approximating graphs cause no ambiguity. This is the standard uniform-root convention of Aldous and Lyons [2],§2, p. 10.

Let P_3 be the path 0–1–2. Write $[P_3, r]$ for its rooted isomorphism class, and define the probability measure

$$\mu = \frac{1}{3}\sum_{r=0}^{2} \delta_{[P_3,r]}.$$

Its two distinct rooted classes have masses 2/3 (an endpoint root) and 1/3 (the middle root).

Lemma 1 (A genuine local weak limit)**.** *Let G_m be the disjoint union of $m+1$ copies of P_3. Then $|V(G_m)| = 3(m+1) \to \infty$, and the uniformly rooted laws of G_m are exactly μ for every $m \geq 0$. In particular they converge locally weakly to μ.*

Proof. Use the actual vertex set $\{0, \dots, m\} \times \{0, 1, 2\}$, with adjacency precisely when the first coordinates agree and the second coordinates are adjacent in P_3. The component of (i, r) is the ith copy, and its rooted isomorphism class is $[P_3, r]$. Each second coordinate occurs $m+1$ times among the $3(m+1)$ vertices. Consequently the pushforward of the uniform vertex law is exactly the displayed measure. Every rooted ball cylinder therefore has the same probability for every m; this is the standard cylinder criterion for local weak convergence. □

A separating cylinder event. For a finite rooted graph (H, o), let $C_r(H, o)$ be the event that the rooted radius-r ball is isomorphic to that of (H, o). Set

$$A = \bigcup_{r=0}^{2} C_2(P_3, r).$$

This is a finite union of genuine local cylinders, hence measurable, and $\mu(A) = 1$.

Lemma 2 (No vertex-transitive graph belongs to A)**.** *For every countable connected locally finite vertex-transitive graph (H, o), one has $[H, o] \notin A$.*

Proof. All vertices of a vertex-transitive locally finite graph have the same degree, because any automorphism induces a bijection between the corresponding neighbor sets. Suppose the rooted radius-two ball at o is a rooted copy of P_3. Every neighbor of o, and every neighbor of such a neighbor, belongs to this radius-two ball. Therefore the degree of the root and the degree of each of its neighbors can both be read from the induced ball: neither neighbor set is truncated.

If the root corresponds to an endpoint of P_3, its degree is 1 and its neighbor's degree is 2. If it corresponds to the middle vertex, its degree is 2 and each neighbor's degree is 1. Either case contradicts equality of degrees in a vertex-transitive graph. This argument does not assert that a radius-two P_3 ball identifies the whole component; for example an endpoint of P_4 has such a ball. □

Theorem 1 (Failure of the claimed decomposition)**.** *There are no mixing probability measure η and measurable family of probability measures (ν_i) on $\mathcal{G}_*$ such that*

$$\mu(B) = \int \nu_i(B)\, d\eta(i) \quad \text{for every measurable } B,$$

and ν_i is concentrated on vertex-transitive graphs for η-almost every i.

Proof. By the separating-cylinder lemma, concentration on vertex-transitive graphs implies $\nu_i(A) = 0$. Substituting the measurable event A in the barycentric identity gives

$$1 = \mu(A) = \int \nu_i(A)\, d\eta(i) = 0,$$

a contradiction. The argument even allows arbitrary measures for the components and the mixing measure, whenever the displayed identity and the almost-everywhere concentration condition make sense. In particular it excludes the conjectured, more restrictive decomposition into extreme probability laws supported on transitive graphs. □

Formalization and verification. The accompanying `lean/Main.lean` uses actual Mathlib `SimpleGraph`s, walk-defined induced balls, root-preserving graph isomorphisms and their quotient, the cylinder-generated measurable space, canonical measures of probability mass functions, and the Lebesgue integral of nonnegative extended real functions. The finite graph component is the induced subgraph on actual reachable vertices; its explicit rooted isomorphism to P_3 is proved. The local law is the pushforward of the uniform PMF on all $3(m+1)$ actual vertices, and equality to μ is proved for every m.

The principal checked results are `copies_local_weak_limit`, `vertex_count_unbounded`, `obstruction_excludes_transitive`, and `no_transitive_barycentric_decomposition`. No assertion that μ is extreme is required for the disproof, and the proof does not challenge the usual ergodic decomposition theorem without the false transitivity restriction.

The independent `verify.py` enumerates every simple graph on at most six vertices, checks the radius-two degree obstruction at every root, checks the P_4 boundary case, and computes the uniform rooted laws of the first forty disjoint-union examples with exact rational arithmetic. These finite computations are consistency checks; the arbitrary-graph and arbitrary-mixture statements are proved by Lean.

Lean 4.19.0 and Mathlib revision `c44e0c8ee63ca166450922a373c7409c5d26b00b` are pinned. In `lean/`, run:

```
lake exe cache get
lake build
lake env lean -DwarningAsError=true Main.lean
```

Run `python verify.py` at the submission root. Compile this paper with `tectonic main.tex`. Actual fresh-build logs and hashes are included. Printed theorem dependencies contain only standard foundational axioms. The work received author, parent and internal cross-review; no external independent review is claimed.

References

[1] The Last Math Competition, Conjecture 00000009305. Exact bilingual source included.

[2] D. Aldous and R. Lyons, *Processes on Unimodular Random Networks*, Electronic Journal of Probability 12 (2007), 1454–1508, arXiv:math/0603062, §2.